

An Analysis of UPI's Security Risks and Remediation Strategies

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Abstract—"The future is cashless", in compliance with this statement, the past decade has seen the successful rise in cashless transactions, owing to an architecture which has made the process seamless and efficient, which is Unified Payment Interface (UPI). UPI is a homegrown technology developed by the National Payment Corporation of India to allow real-time transactions. Pedaled by the digitalization in India, UPI has been increasingly adopted by the people from all walks of life – from a street vendor to a multinational corporation. As of 2023, the number of payments made through UPI reached a whopping 117.6 billion worth Rs. 182 lakh crores. Primarily designed for mobile users, the UPI technology has made it possible to perform transactions at the tip of your fingertips, providing 24/7 * 365 days availability. One of the major advantages of UPI lies within its interoperability feature, where the sender can transfer funds to the receiver having accounts belonging to different banks. UPI has also helped in inhibiting corruption in a developing country like India. The UPI infrastructure has been implemented on a national level, making it prone to vulnerabilities in its security in spite of mechanisms to prevent being targeted by malicious entities. These mechanisms include encryption protocols, authentication methods, data protection measures and fraud prevention techniques. As the scale of application of UPI grows, so does the requirement for its security to be enhanced (from a squalor to a fortress). These security measures must make the UPI framework impenetrable against cyber crimes.

Index Terms—UPI, Security, Social Engineering, Cyber crimes,

I. INTRODUCTION

UPI is a homegrown, cashless payment technology that has become the Goliath in the Indian Transaction landscape. From its pilot launch in the year 2016 to present, UPI transactions have surged a whopping 5000 per cent. The transactions have grown from just 92 crore in FY-2017-18 to a massive 8,375 crore in FY 2022-2023.[1] Among all the countries in the world who have adopted different digital transaction technologies, the country with the highest digital transactions is India, accounting for nearly 46 per cent of all total digital transactions around the world.[7]

There are plethora of reasons that led to rise of UPI. Interoperability being one of the most important reasons, that is, all QR codes are compatible with the CBCD apps. UPI combines several financial functions by connecting numerous bank accounts into a single smartphone application.

Other reasons that lead to the adoption of UPI on such a wider scale was the robustness of the digital payment infrastructure and a slight behavioural nudge which incentivised switching to cashless form of payment. This behavioural change were cemented by creating trust and making it accessible to the core group.[7] UPI has also overall helped in the financial inclusion and contributed in the economic development of the nation[8].

For a successful transaction on a UPI platform, the sender and the recipient need to go through the onboarding process. During this process, they need to make sure that their mobile numbers are connected to their bank accounts on the device that initiated the transaction for authentication. Establishing a Virtual Payment Address (VPA) expedites money transactions by providing a unique identity for each party. The Payment System Player (PSP) receives a payment request from the payer, which is subsequently verified at both the PSP and NPCI levels using a private key. Banks, NPCI, and PSPs exchange confirmation messages while NPCI acts as the central processing point, debiting monies from the payer's account and crediting them to the payee's account. The PSP notifies the payer that the transaction was successful after it is finished.[11]

As UPI gets widely adopted, it gets equally important to strengthen the security measures and mitigate the risks of security breach. As hackers tend to be highly motivated when it comes to money, it becomes essential to discover any design vulnerabilities and address them.[5] The previous protocol UPI 1.0 revealed significant design weaknesses in its multi-factor authentication workflow, potentially leaving users vulnerable to attacks facilitated by attacker-controlled apps.[11] These vulnerabilities, including unauthorized registration and transactions using user cell numbers and partial debit card information, pose serious threats to the security of transactions conducted via UPI.[11] The introduction of UPI 2.0 protocol helped in tackling the mentioned attack vectors. UPI 2.0 also added steps that focused on strengthening authentication, encryption, and transaction validation procedures, hence improving overall security against fraud and unauthorized access. This paper focuses on pre-existing security measures in different UPI payment platforms, discusses possible vulnerabilities

and solutions.

II. LITERATURE REVIEW

Unified Payment Interface has amassed over 11768 crore payments [10] through its easy-to-use, mobile-only interface [3]. The working of UPI lies in the extensive use of a Virtual Payment Address (VPA). Virtual Payment Address uniquely identifies a user digitally by using a translation service and mapping it to their bank account. This virtual address effectively hides the financial identity of the customer [12]. A VPA is one of the most significant changes that the world of online payment saw as it removes the requirement for additional bank information, like account numbers and IFSC codes, to be disclosed during transactions, which is the case during regular online transactions such as NEFT, Credit and debit transfers. But even so, the world of UPI transactions is not completely impenetrable. Despite several issues and weaknesses, the government and banks continue to support online and mobile banking due to these platforms' accessibility, user-friendliness, mobility, and capacity to serve a large user base.

The purpose of this review is to go over the existing security measures implemented in UPI and how they were still manipulated to conduct thoroughly planned cyber-attacks.

The pillars of security implemented in UPI are as follows:

- Multi factor authentication, in the form of the initial 4-digit code entered by the user to open the application of choice: GPay, PhonePe, PayTM, BHIM etc and the second layer being the 6-digit MPIN that is associated with an individual account.
- Multi layered encryption: Advanced encryption protocols like TLS (Transport Layer Security), AES (Advanced Encryption Standard), and PKI (Public Key Infrastructure) [9] safeguard Multi-Layer Encryption UPI transactions, guaranteeing the confidentiality of sensitive data during transmission, including user credentials and transaction data.
- Device Binding: In the context of UPI, device binding refers to linking a specific mobile device or application to a user's UPI account credentials. When a user registers for UPI, they typically link their mobile number to their bank account(s) through a UPI-enabled mobile application. This process involves securing the account by verifying the mobile number and linking it to the bank account(s). This adds an additional layer of security and helps users maintain control over their UPI account by restricting transactions only to the registered device.
- Reliance on the Push-Pull transaction architecture – UPI provides two types of transactions – Push and Pull. This architecture ensures the safe and accurate peer-to-peer payment between the already verified and authenticated users.
 - When doing a push transaction, the payer first enters the payee's VPA address before starting the transaction after which, the request is sent to the UPI servers, which use their database to search for the

payer's and the payee's bank accounts. The money is then debited from the payer's account and credited to the payee's account. In this manner, the transaction is finished and both the payer and the payee are made aware of it.

- In a push payment, the payee starts the transaction by providing the payer's VPA address. The payer receives a request for payment from the payee after the PSP forwards the request to the UPI servers, which forward it to the payer's PSP. He has the option to approve or reject the request. If he grants the request, he inputs his MPIN and the transaction is completed; both parties receive a notification that the transaction was completed successfully once the funds were successfully transferred from the payer to the payee.[6]

Although the above-mentioned measures have largely ensured efficient transactions, UPI has seen a few breaches in its security. According to government data, there were 84,000 cases of UPI fraud in 2021-22, and in 2020-21, 77,000 such cases were recorded. According to the union finance ministry, more than 95,000 fraud cases of UPI transactions were recorded in the country in 2022-23 [9]. These cases can be categorized into two categories based on their broad reason for occurrence:

- Use of social engineering to mislead users. Social engineering can be summed up as using human behavior to circumvent security without the target (or participant) being aware of it[2] and it is the “psychological manipulation of an individual or set of individuals to produce a desired effect on their behavior”[13].
 - Typically, frauds include a hacker employing UPI QR codes to fool a victim or merchant into divulging their UPI credentials—such as their UPI ID or PIN—through social engineering.[3]
 - Watering Hole attacks: These attacks typically target merchants and heavily rely on social engineering tactics.[3]
 - * Attack Process: The attacker identifies the merchant's shop and determines the UPI Scanner in use. Using manipulation, often posing as a UPI payments company technician, the attacker obtains the QR scanner from the merchant.
 - * QR Code Replication: The attacker replicates the QR code scanner exactly like the merchant's, replacing only the QR code with their own.
 - * Payment Redirection: This manipulation allows payments intended for the merchant to be redirected to the attacker's account.
 - * Detection Challenges: Merchants find it challenging to detect this scam due to the unique and unreadable nature of the UPI name assigned to them. The UPI ID consists of a series of random characters that are difficult to memorize.
 - * Exploiting UPI Name: Exploiting the complexity

of the UPI name, the attacker successfully implements the social engineering attack, deceiving merchants engaged in UPI-based payments.

- Using the underlying UPI app or device infrastructure: Attackers may use a rooted phone and use many tools to reverse-engineer the payment apps. [5] Reverse engineering involves analyzing the inner workings of a software application or system to understand its functionality. Rooted phones are those which have undergone a process of “rooting” which enables their users to access system files and settings that are typically restricted. Further, attackers may use methods to de-compile the binary code of the payment app to a human-readable format, making the source code visible and thus putting the application at risk of being misused. Attackers also constantly monitor the users’ activity of the payment application by analyzing the API calls made by them and may inject malicious code or manipulate the existing code to serve their needs.

As the use of UPI leaps beyond limits, it becomes of paramount importance to focus on increasing security measures to battle such inconsistencies. The underlying goal here is to minimize the attacks occurring through social engineering by increasing awareness about digital literacy and to conduct manual verification after each payment on any of the UPI applications. Usage of verifying devices like the PayTM soundbox will also help merchants in confirming that a payment took place successfully. The paper will further discuss the implemented security measures and provide plausible measures to battle the latest fraud attacks occurring in UPI.

III. METHODOLOGY

This paper uses the existing literature to understand the vulnerabilities, the security measures in-place and the steps which can be taken to improve the security of the digital payment infrastructure. It does so by reviewing the research which uses reverse engineering and a proprietary protocol, to learn about UPI’s authentication handshake.[5]

- Protocol Analysis: Protocol is a set of well-defined rules and analysis is the thorough examination of each of the rules. In order to understand the vulnerabilities a system and to recreate it, there is a need to understand how UPI does fingerprinting of each device and establish which credentials user requires to set up account and perform transactions. [5]
- Reverse-Engineering: In order to understand the transaction process, and to discover vulnerabilities with the structure of the UPI system it is quite crucial to familiarize with each layer of the system.

The UPI process flow is illustrated by the given figure:

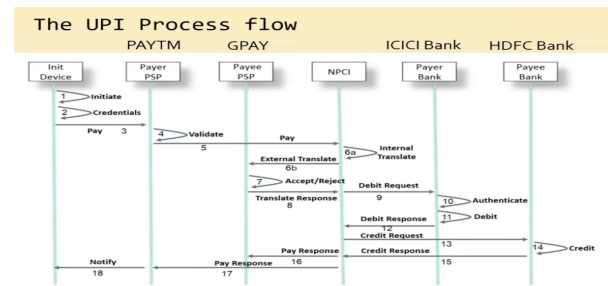


Fig.1. The UPI Process Flow

The methodology used in the referenced research paper has a few limitations. As the research did not have direct access to UPI’s servers, it limits the analysis to reverse-engineering through client-side interactions[5]. The effectiveness of the methodology relies on the availability and functionality of UPI-supporting payment apps which may vary in terms of protocol implementation and security measures. Furthermore, there are apps like GPay, which may employ methods preventing repackaging limiting the ability to instrument and analyze their code[5].

In addition to this, statistics and other information provided by the governing body of UPI, National Payments Corporation of India (NPCI) are used along with articles from independent organisations for this research paper.

IV. UPI AND ITS ROBUSTNESS

UPI observes a large demographic of users all over India. According to Dev Harshal et. al. discuss in their paper, just 7% of survey participants said they had cut back on spending as a result of UPI, compared to over 75% who said they had increased it. Notably, 91.5% of respondents said they were happy with the way they used UPI. Additionally, 95.2% of survey participants said that using UPI to make payments was convenient. This is mainly owing to its secure two-factor authentication, multi layered encryption and data binding features being implemented in these UPI apps. However, despite the implementation of these features, falling victim to a financial crime on UPI is easier than meets the eye. As appearing on I4C’s quarterly newsletter, ‘Cyber Pravah’, some statistical facts for Q2 2022 were as follows:

- Financial institutions withheld transactions under the Citizen Financial Cyber Frauds Reporting and Management System (CFCFRMS) worth about 34 crore rupees.
- 4,953 forensic services were provided by the National Cyber Crime Forensic Laboratory (NCFL) to LEAs in states and UTs.
- 12,382 mobile numbers were linked to cybercrime cases across India.

Such elaborate crimes have occurred not due to the incompetency of UPI or its underlying architecture, but because malicious attackers have targeted the most basic

fundamental network protocols and used tricks infamous in the field of social engineering to scam people out of their hard-earned money. Below listed are some of the potential risks associated with UPI that attackers have previously made use of to conduct cyber crimes:

- **Device Binding** : As discussed earlier in the literature review, device binding is a mechanism that has helped secure UPI transactions by linking an individual's bank account to their registered mobile number to add a layer of security. However, the same mechanism can be seen as a double-edged sword, as discussed by R. Kumar et. Al [2], A user or an adversary can attach her phone to a cell number linked to another user's bank account through a weak device binding technique.
- **Spear Phishing** : A spear phishing assault is an attempt to gain access to a computer system or sensitive information by sending phony messages that look real. [3] These emails are sent to the target with the intention of persuading them to click on a malicious link or attachment, which will expose them to malicious software. Phishing emails are careful to manipulate users with valuable information, such as CEOs, or on the other hand, certain inexperienced employees. This type of attack makes up a major part of cyber attacks, owing their successes for the following reasons:
 - 1. **Instilling Fear and urgency** : Attackers draft phishing emails injected with malicious software by prefacing them with information to create a false sense of urgency, urging the target to immediately send confidential information, threatening compromise to their systems.
 - 2. **Persuasive brand e-mails** : Expert hackers create emails that seem real by imitating well-known companies like Microsoft, Apple, or the user's bank.
 - 3. **Insufficient training for staff members**: Without ongoing instruction, staff members may be unable to detect or report spear phishing attacks, leaving them vulnerable targets.
 - 4. **Generative artificial intelligence (AI)**: Scammers nowadays create convincing emails that imitate human communication styles by using AI technologies such as ChatGPT, which makes their phishing schemes even more difficult to uncover.

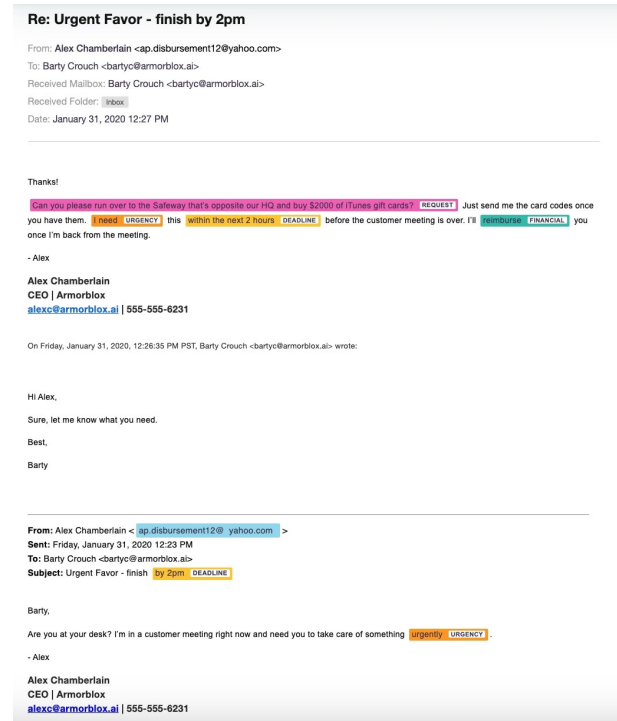


Fig.2. An example phishing email

- **Ransomware** : Ransomware extorts a victim's data or device, threatening to lock it or worse, unless the victim pays the attacker a ransom. Ransomware attackers use sophisticated encryption techniques to hijack victims' files in exchange for a large ransom to release encrypted data[4]. Ransomware attacks may reach the intended targets by a phishing e-mail, for example, the ransomware might come disguised as a harmless looking .pdf, Microsoft Word document, or other file. Ransomware may come in different forms. ransomware may be encrypted or non-encrypted. While some ransomware e-mails achieve their intended goals only through a message that is meant to scare the targets, i.e, scareware, other forms of these malware may be achieved through active, prolonged threatening, where the attackers obtain the personal information of the target, who may be individuals or organizations and leverage a sum of crucial information in order to trick the targets into conceding. These methods come under encrypted ransomware. The lesser known type of ransomware, or the non-encrypted type screen-locking ransomware, locks the victim's entire device, usually by blocking access to the operating system. Instead of starting up as usual, the device displays a screen that makes the ransom demand.[14]
- **Botnets** - A botnet, short for "robot network," is a collection of malware-infected computers that are managed by a single attacker known as the "bot-herder.". This is a centralized type of cyber-attack, where the herder can simultaneously gain access of multiple targets

through commandeering the individual bots. Financial breaches are one of the major prey of botnets, the most prominent one being the ZeuS botnet which has been responsible for attacks involving millions of dollars stolen directly from multiple enterprises over very short periods of time.

What is a Botnet ?

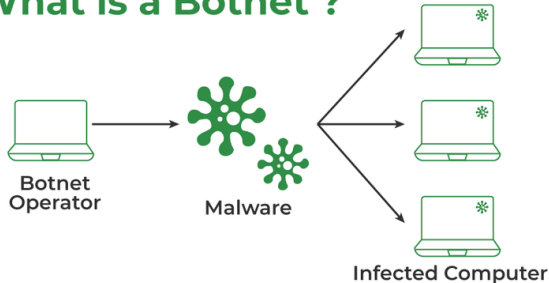


Fig.3. Working of Botnets

- **Whaling** : A specific type of phishing assault called a "whaling attack" aims to obtain sensitive information from a company by emailing key employees. A common goal of these attacks is to convince the target to approve high-value financial transactions to the attacker. Although cloud storage and file hosting sites, online services, and e-commerce sites are getting a higher percentage of attacks, financial institutions and payment services are the companies that are attacked the most.
- **Money Mules** : A money mule is a someone who obtains and transfers funds obtained from deceived parties. While some money mules are aware that their acts are supporting fraudsters, others are not aware that they are aiding criminal behavior. Fraud rings transfer money to an intermediary account to store the loot once they have your financial information. This account serves as one of the money mules used to store funds obtained from several gullible victims. This kind of deception is frequently encountered in UPI hacking.

V. AN EXAMINATION OF THE CURRENT DEFENSIVE TECHNOLOGICAL INFRASTRUCTURE AND ITS INHERENT VULNERABILITIES

Unified Payment Interface has simplified many aspects of the current Indian's daily life. From grocery shopping to cab payments, from inter-account transfers to bill payments like – mobile recharging, gas bill payment, TV subscriptions, UPI has become the face of digital payments with apps like GooglePay, PhonePe, PayTM being an essential instalment in almost every Indian's smartphone. According to the NCPI, UPI started with setting up of vision of 'becoming best global payment network'. This level of large-scale and every-growing success is owed to its highly efficient and capable security measures implemented in UPI. As mentioned in the literature review, UPI implements many security measures

such as multi-factor authentication, data encryption and device binding to ensure security of its transactions. Since it relies heavily on a monumental amount of bank accounts linked to individual users' accounts, UPI can be classified as a type of biometrically authorized payment structure. Over 99 percent of Indian individuals have their biometric characteristics digitally recorded in the country's "Aadhaar" database. In terms of regulations, a biometrically-authenticated payment system consists of three components. Every customer's bank account and Aadhaar data are connected at the back end. The system appears as a "biometric capture" device at the frontend, using the user's Aadhar credentials to verify the user and to further, link his/her bank account to the account. The system itself is made up of a network for data transfer between the two ends. In addition to the already existing security measures implemented in the back-end, below listed are some of the other measures:

- **Tokenization**: UPI utilizes tokenization to replace sensitive card or account information with unique tokens. These tokens are meaningless to attackers even if intercepted, thus mitigating the risk of data breaches and identity theft.
- **Risk-based Monitoring**: UPI platforms employ sophisticated algorithms to continuously monitor transactions for suspicious activities. Any unusual patterns or deviations from normal user behavior trigger alerts, allowing for prompt intervention and mitigation of potential threats.
- **Secure APIs**: UPI interfaces with banks and third-party service providers through secure Application Programming Interfaces (APIs). These APIs adhere to industry-standard security protocols and best practices to ensure the integrity and confidentiality of data exchanged between different entities within the payment ecosystem.
- **Regular Security Audits**: UPI undergoes regular security audits and assessments conducted by independent third-party security firms. These audits help identify vulnerabilities and weaknesses in the system, allowing for timely remediation and strengthening of security measures.
- **Customer Education and Awareness**: UPI providers prioritize customer education and awareness campaigns to educate users about safe banking practices, common scams, and security best practices. This empowers users to recognize potential threats and take proactive measures to safeguard their accounts and transactions.
- **Introducing Re-KYC in cybercrime hotspots**: Investment and financial services firms utilize Know Your Client (KYC) to confirm the identification of their clients and any potential hazards related to the client connection. Re KYC is a measure taken by banks to guarantee that the details obtained at the time of bank account establishment, such as KYC documents, contact information, address and so on, are up to date.
- **Capturing geo-location during transactions**: Integrating geo-location tagged based services would add an added layer of security. Firstly, geo-location data can be used

as one of the parameters to detect potentially fraudulent transactions. If a transaction originates from an unexpected or suspicious location that deviates from the user's usual patterns, it may trigger an alert for further verification or scrutiny. Also, it enables prompt detection of anomalies or suspicious activities by providing real-time visibility into the geographic distribution of transactions.

- Cyber tipline reporting system: Steps have been taken to improve the system of filing cyber fraud and related grievances via Cyber Tipline Reports and their retention on its database. This software was adopted from the National Cybercrime Police Portal (NCP) of the National Centre for Missing and Exploited Children (NCMEC) in USA.

VI. EXISTING VULNERABILITIES

- A central database is where Aadhaar data are kept on the back end. It is a well known fact that centralizing the storage of sensitive data is a poor security practice, and there are several examples of the dangers that come with it.
- In any default or alternate handshake, the UPI protocol provides access to a user's bank account details provided they have their mobile number and no bank-related credentials.
- The first factor, device hard-binding, depends on data that may be readily extracted from a device. A user (or an adversary) can attach her phone to a cell number linked to another user's bank account thanks to a weak device binding method.
- UPI does not require the user to supply printed debit card information or any bank credentials in order to approve transactions from a new phone when an existing user transfers their UPI account to a new phone. It is fully reliant on the UPI PIN alone.
- The partially disclosed debit card information printed on the card is necessary to set the second factor, the UPI PIN, and therefore is not private. A user's shared secret with the bank, their debit card PIN, is never used. This is less strict because both in-person and online transactions demand the whole card number and PIN from the debit card.

VII. POSSIBLE SOLUTIONS

- Biometric Authentication: Integration of advanced biometric authentication methods such as facial recognition or voice recognition can add an additional layer of security to UPI transactions. Biometric data is unique to each individual and difficult to replicate, making it a robust authentication factor.
- Behavioural Biometrics: Leveraging behavioural biometrics, such as typing patterns, touchscreen interactions, and device orientation, can help detect anomalies and unauthorized access attempts. Analysing user behaviour in real-time can enhance fraud detection capabilities and improve the overall security of UPI transactions.

- Blockchain Technology: Incorporating blockchain technology into the UPI infrastructure can enhance security by providing a tamper-proof and transparent ledger of transactions. Blockchain ensures data integrity and immutability, reducing the risk of data manipulation and fraud within the payment ecosystem.
- AI-Powered Fraud Detection: Leveraging artificial intelligence (AI) and machine learning (ML) algorithms for proactive fraud detection and prevention can improve the effectiveness of security measures in UPI. AI-driven fraud detection systems can analyze vast amounts of transaction data to identify patterns, anomalies, and emerging threats with greater accuracy and speed.
- Improvements to the Regulatory Framework: UPI providers can be guaranteed to comply with the strictest security and compliance requirements by means of ongoing modifications and improvements to the regulatory frameworks that oversee digital payments and financial services. Regulatory agencies and industry participants should work together to create standards and best practices for improving UPI security and successfully fending off new threats.

CONCLUSION

In conclusion, the security measures implemented in the Unified Payments Interface (UPI) represent a comprehensive approach to safeguarding users' financial transactions and personal information in the digital realm. Through a combination of encryption, authentication protocols, real-time monitoring, and regulatory compliance, UPI providers strive to create a robust ecosystem that mitigates risks associated with fraud, unauthorized access, and data breaches. As the landscape of digital payments continues to evolve, ongoing investments in advanced technologies, collaborative efforts among stakeholders, and adherence to stringent security standards will be essential to maintaining the integrity and trustworthiness of UPI as a secure and convenient payment platform for millions of users across India and beyond.

The interoperability of UPI, which enables customers to conduct business with several banks and payment service providers without difficulty, is the key to its success. Interoperability should be given top priority by practitioners in their fintech solutions in order to improve user experience and market reach. The strong security features of UPI demonstrate how crucial it is to give security first priority while developing fintech goods and services. It is recommended that practitioners make investments in sophisticated encryption, authentication protocols, and fraud detection systems to safeguard users' personal data and financial resources. The great acceptance of UPI can be attributed in part to its user-friendly design. It is recommended that practitioners give priority to user-centric design concepts in order to develop fintech solutions that are easy to use and meet a range of user needs and preferences.

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